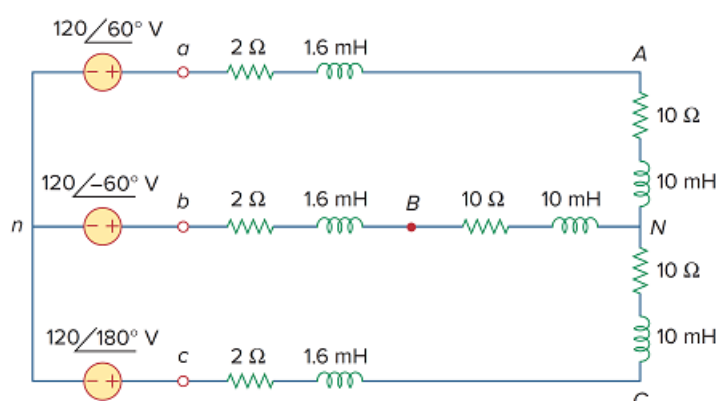


Problem

Refer to the balanced Y-Y circuit of Fig. 12.29. Use *PSpice* to find the line current I_{bB} and the phase voltage V_{AN} . Take $f = 100$ Hz.

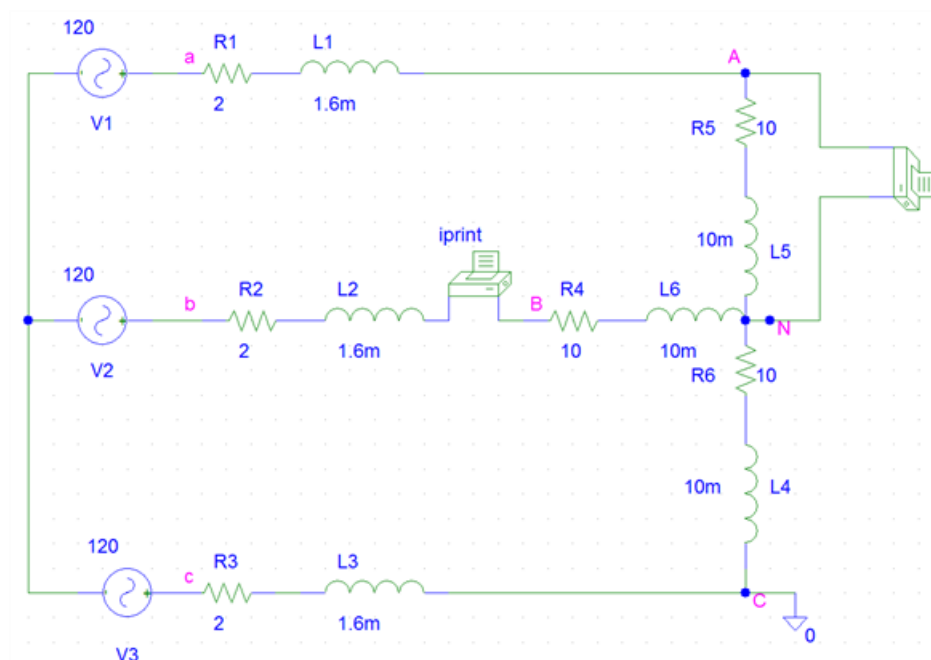
Figure 12.29



Step-by-step solution

Step 1 of 3

The schematic is shown in below



Double click on 'VAC' voltage source and set AC MAG and AC PHASE for given values. The pseudo components *iprint* is inserted in the appropriate lines to obtain I_{bB} , while *VPRINT2* is inserted between nodes A and N to print differential voltage V_{AN} . We set the attributes of *iprint* and *VPRINT2* each to AC = yes, MAG = yes, PHASE = yes, to print only the magnitude and phase of the currents and voltages.

As a single – frequency analysis. We select analysis/setup/AC sweep and enter total pts =1, start freq = 100 and final freq = 100. Once the circuit is saved, it is simulated by selecting analysis/ simulate. The output file includes the following.

Step 2 of 3

FREQ VM(\$N_0009,\$N_0004) VP(\$N_0009,\$N_0004)

1.000E+02 1.009E+02 6.087E+01

FREQ IM(V_PRINT1) IP(V_PRINT1)

1.000E+02 8.547E+00 -9.127E+01

Step 3 of 3

Therefore

$$V_{AN} = 100.9 \angle 60.87^\circ \text{V} \text{ And}$$

$$I_{bB} = 8.547 \angle -91.27^\circ \text{A}$$